

Topic: Graphing Equations.

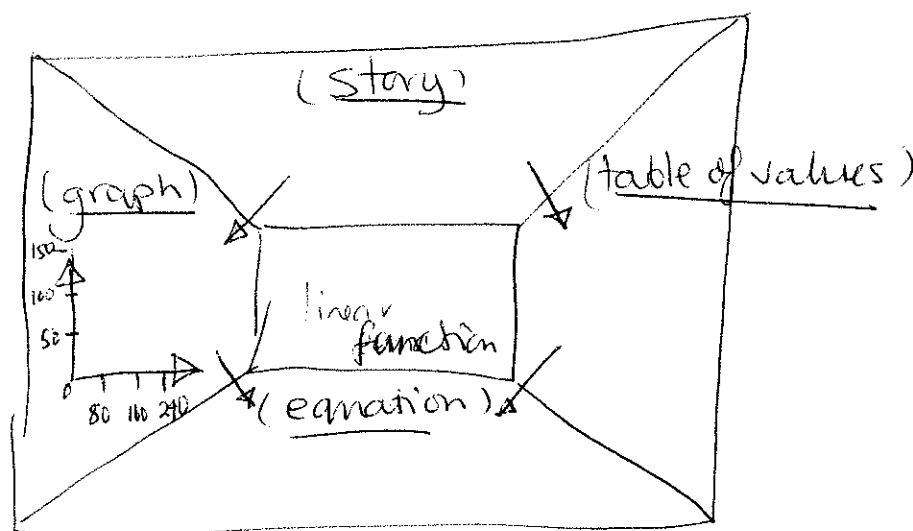
EQ: Can an equation "tell us" what type of graph it will be?

What features can we look for?

Qns

Notes:

Let's use $y = 80x + 50$
and a think board.



What are the main features of an equation?
(gradient + vertical intercept)
of a graph?
(vertical intercept) and shape.

- ① We might begin with a real-world context.
We pay \$50 for a plumber to arrive and \$80 per hour or part hour he takes to do the job.

This is the Story.

- ② The ~~Equation~~ graph suggests that we start at \$50 (not zero) ^{on the y-axis (vertical intercept)} and plot \$80 intervals along the horizontal axis.

- ③ we can ~~now plot~~ ^{also construct} our table of values.

x	0	1	2	3
y	50	130	210	290

By starting at zero for x, and 50 for y ^{as we ~~now~~ know the vertical intercept.}

By increasing our initial \$50 by \$80
 each time we increase x by 1, we
 can see the gradient is 80, by drawing
 ↗ ↗ ↗ arrows and +80 each time.

④ From using either the Table of Values or
 the graph, it is easy to write the equation.

$$y = \boxed{80}x + \boxed{50}$$

↑ ↑
 gradient Vertical Intercept.
 (steady (Initial cost)
 increments)

Summary: This graph is a linear function

so we can write this, inside the board.
 we look for the gradient and the
 constant

$$y = m x + c$$

↑ ↑
 gradient constant

think Taxi-
flagfalls?

Extn:

LHS: Now write a story for your partner to

Think board... + swap.....

or start with the equation?